



Computer Organization & Architecture

Introduction to Memory

“Memory is the faculty of the brain by which data or information is encoded, stored, and retrieved when needed.”

-- Wikipedia



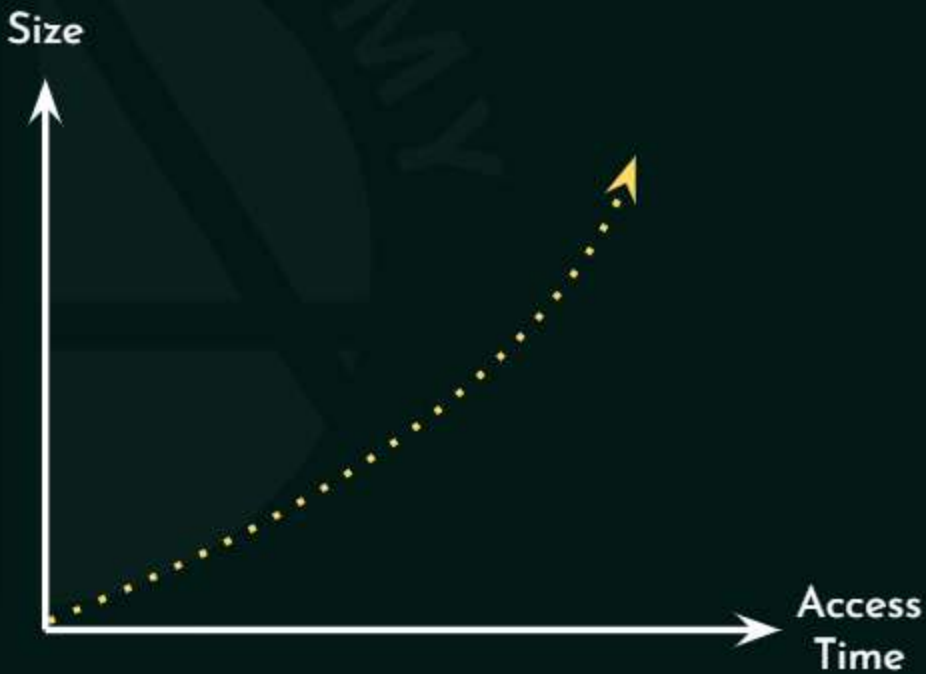
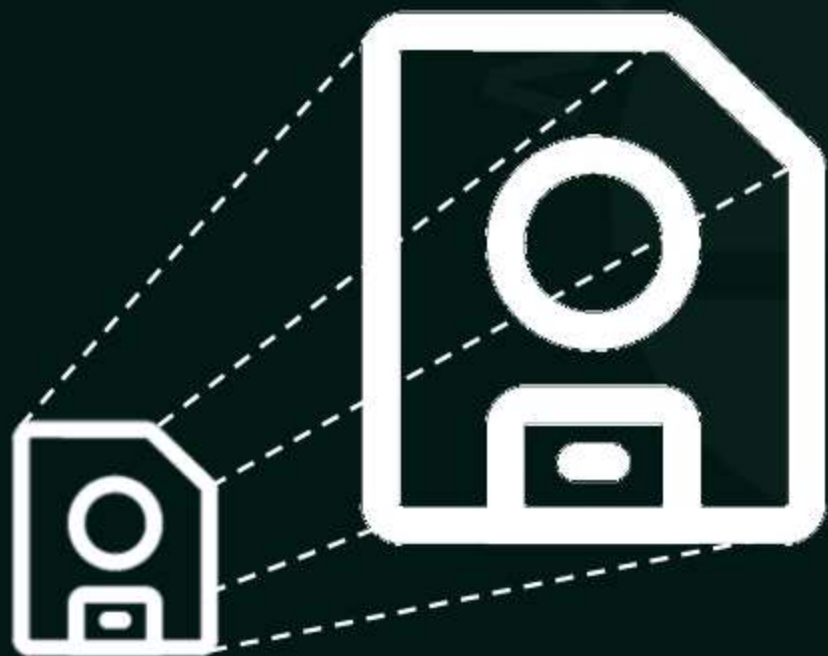
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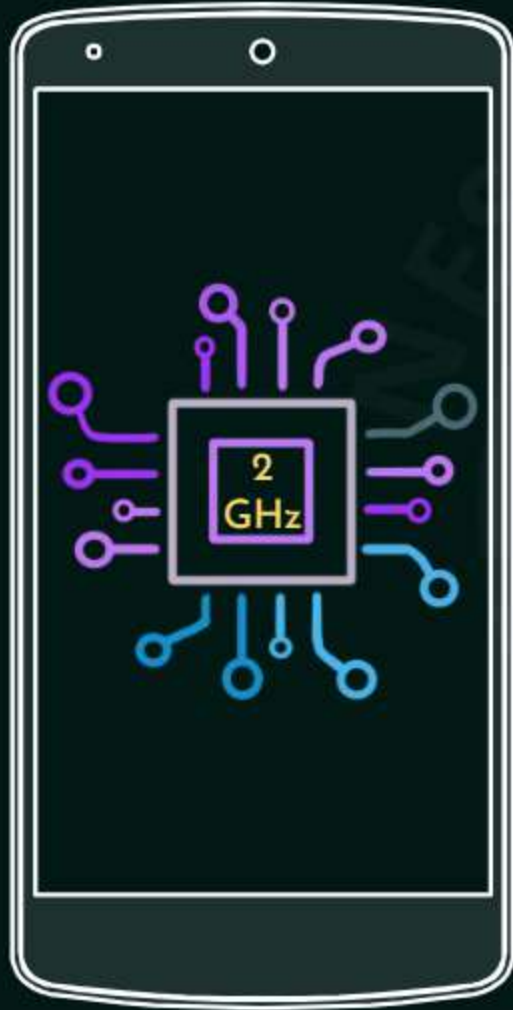
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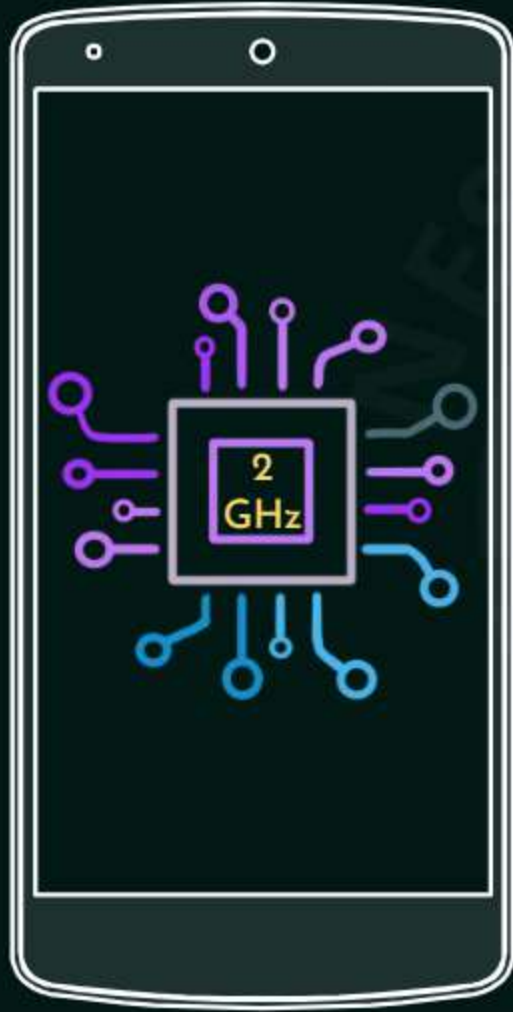




Frequency: 2 GHz

$$\text{Time: } \frac{1}{\text{Frequency}}$$
$$= \frac{1}{2 \times 10^9} \text{ Sec.}$$

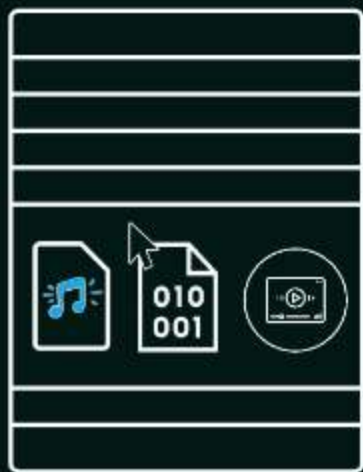
1 Kilo <Unit>	= 1000 <Unit>	= 10^3 <Unit>	
1 Mega <Unit>	= 1000 Kilo <Unit>	= $10^3 \times 10^3$ <Unit>	= 10^6 <Unit>
1 Giga <Unit>	= 1000 Mega <Unit>	= $10^3 \times 10^3 \times 10^3$ <Unit>	= 10^9 <Unit>



Frequency: 2 GHz

$$\begin{aligned}\text{Time: } & \frac{1}{\text{Frequency}} \\ &= \frac{1}{2 \times 10^9} \text{ Sec.} \\ &= \frac{1}{2} \times 10^{-9} \text{ Sec.} \\ &= \frac{1}{2} \text{ nsec.}\end{aligned}$$

Primary Memory



Secondary Memory



Primary
Memory



Dynamic Random Access Memory
D.R.A.M.



Primary Memory



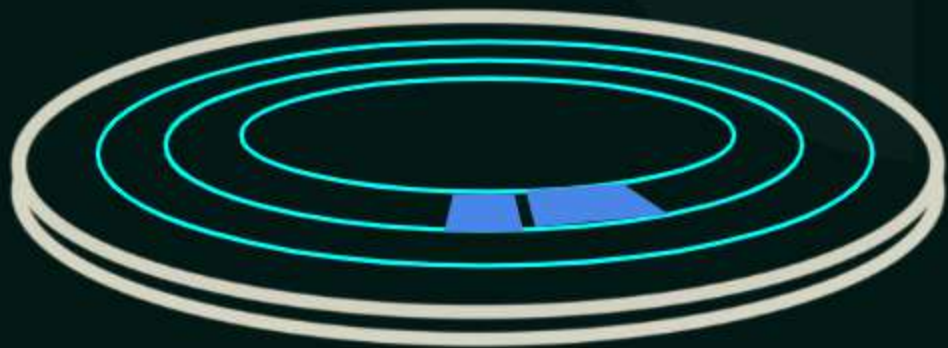
Dynamic Random Access Memory
D.R.A.M.



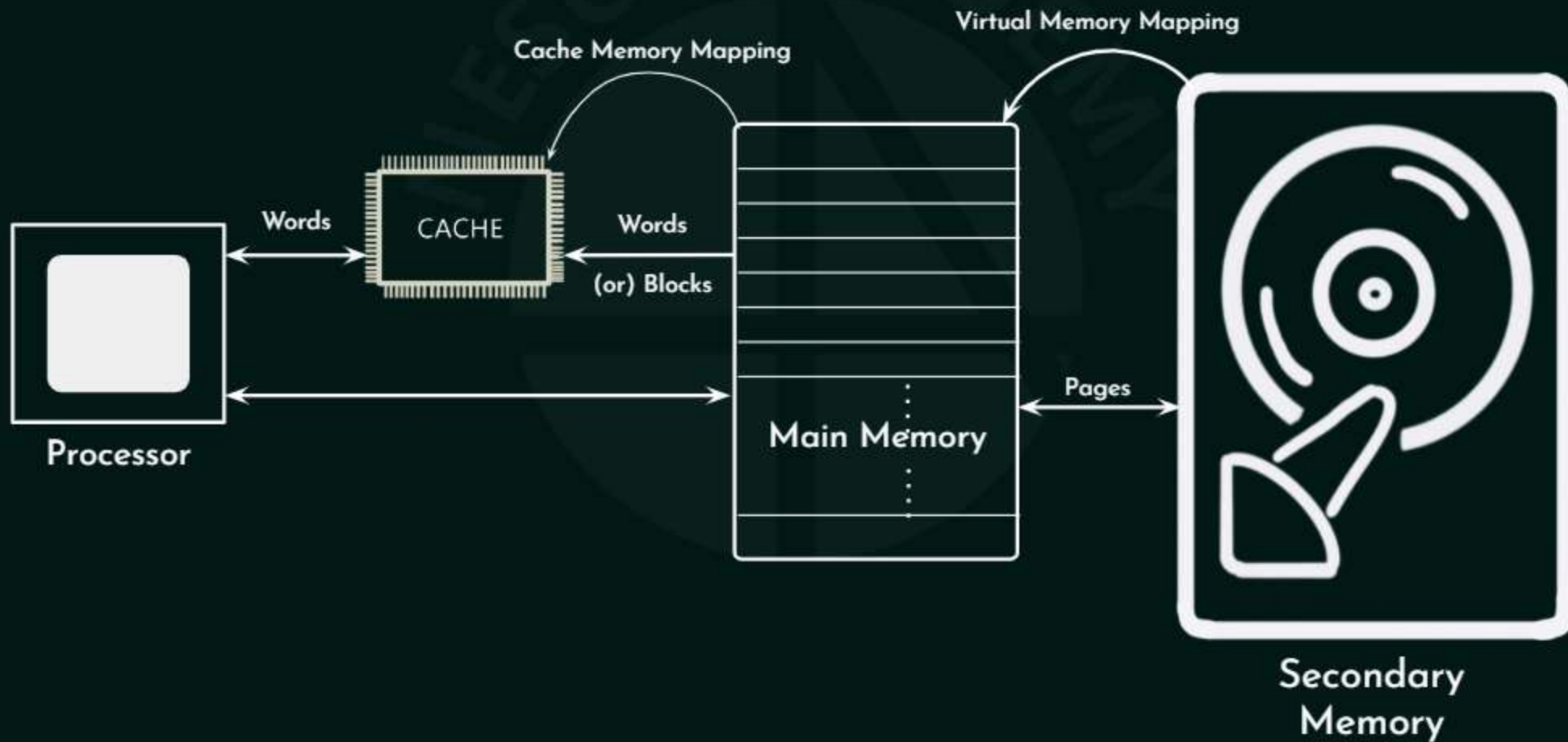
Static Random Access Memory
S.R.A.M.

Secondary Memory :

- **Slower** than Primary Memory.
- Retains Data **Permanently**.
- **Bigger** in size.
- **Cost-effective**.
- **Semi-Random** accessibility.



THE BIG PICTURE





Computer Organization & Architecture

Memory Hierarchy & Interfacing

HIERARCHY



RANKING



Release Year



**Rotten
Tomatoes**



metacritic

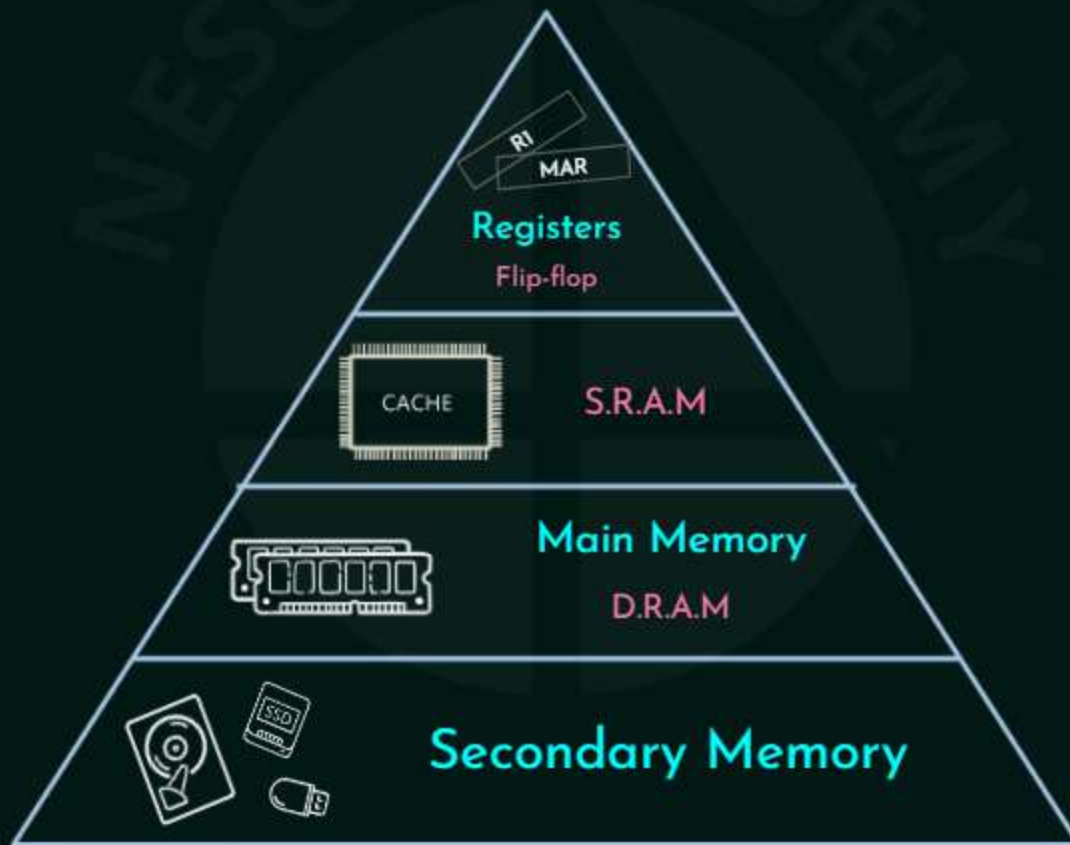
MEMORY HIERARCHY

Access
Time

Size

Cost

Usage
Frequency



Memory Interfacing :

- Part of Computer Organization.
- Deals with the way of connecting various levels of Memory units to Processor & I/O peripherals.



Processor

Unit of measuring speed : MIPS (Million Instruction Per Second)

Memory Interfacing :

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$$n^{\text{th}} \text{ Level} \subseteq (n+1)^{\text{th}} \text{ Level}$$



Processor

n^{th} level

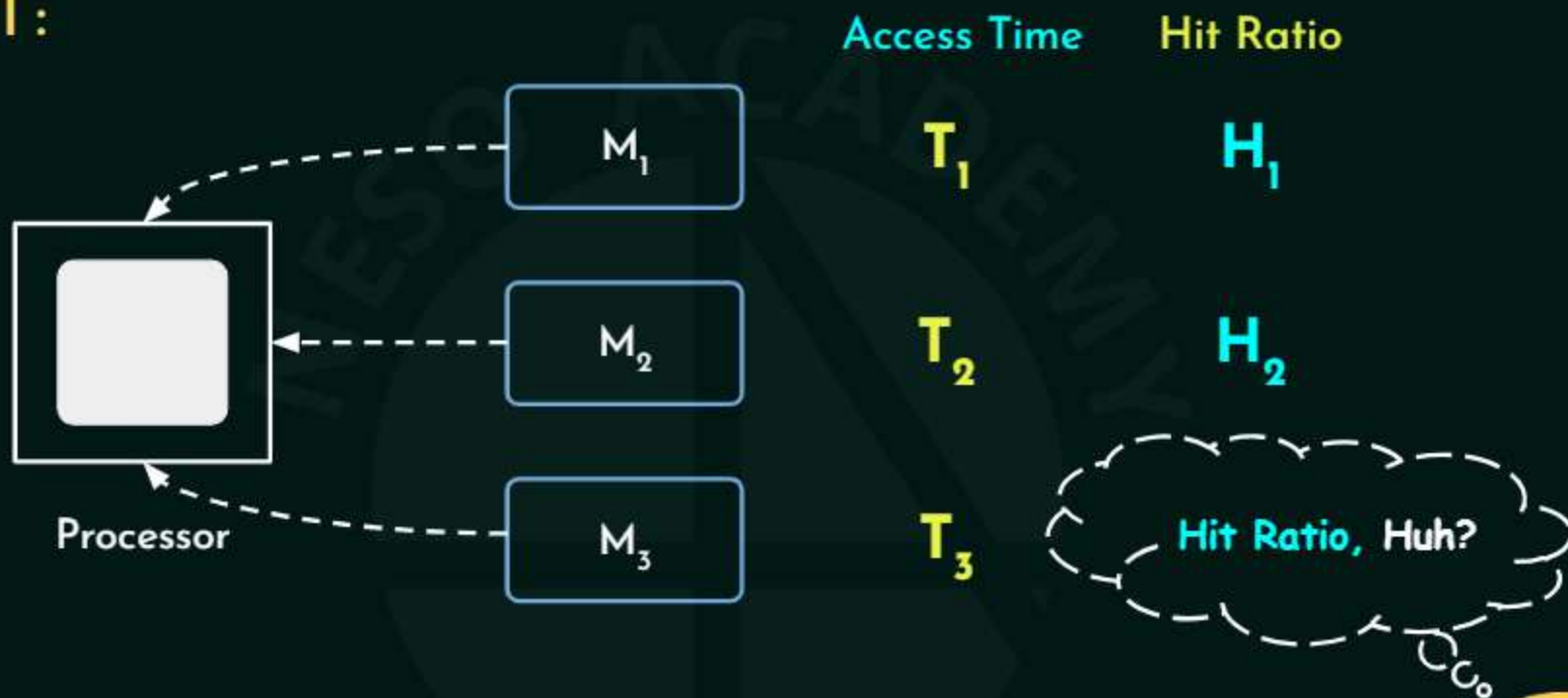
$(n+1)^{\text{th}}$
level

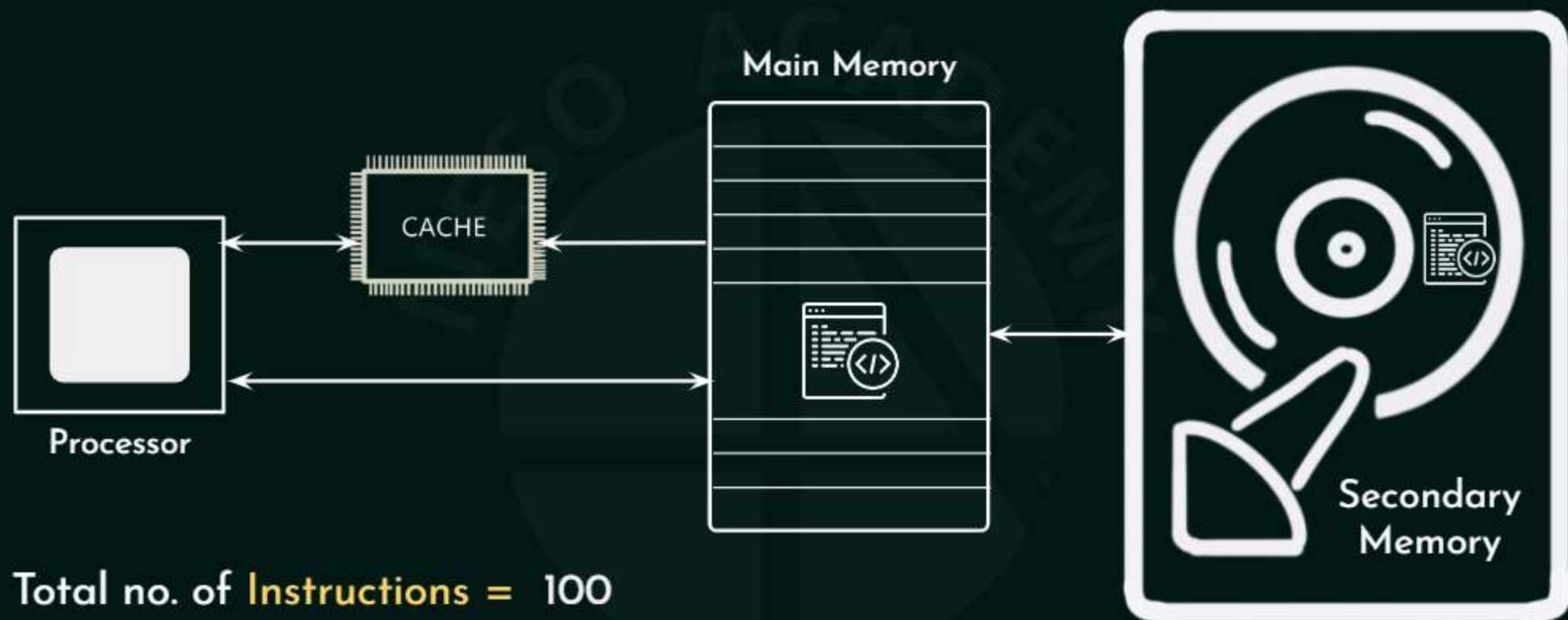
✓ Hit

✗ Miss



Way 1:





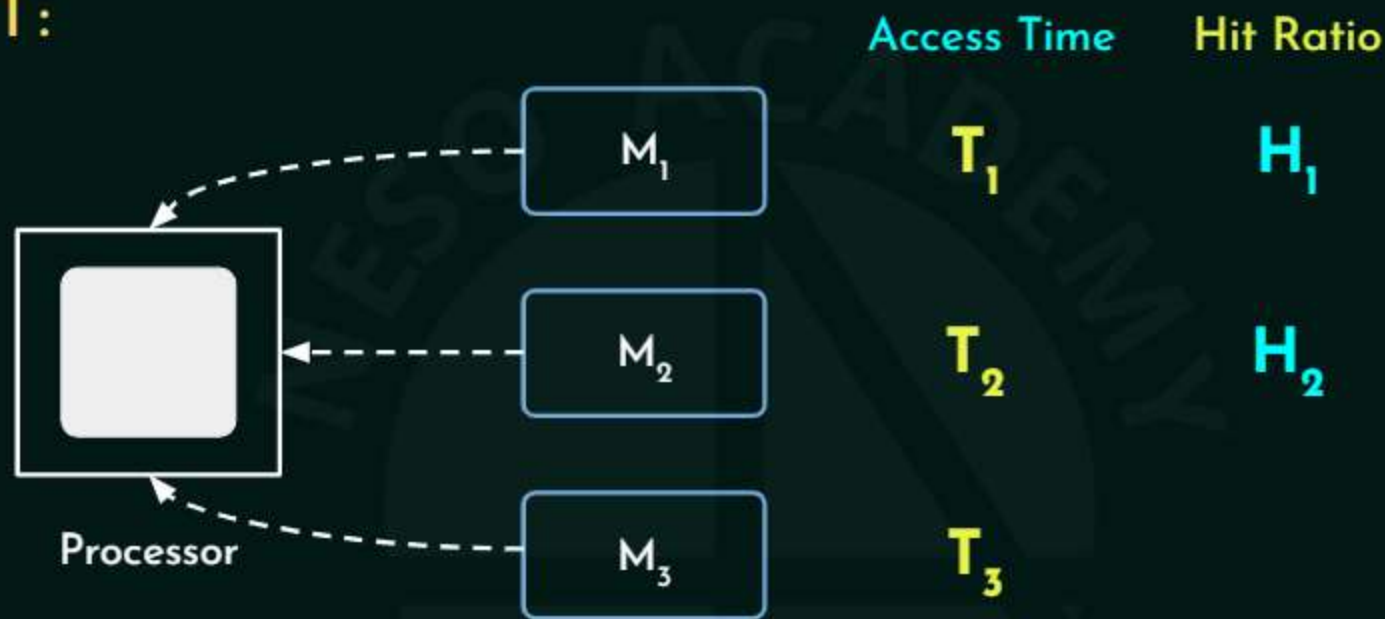
Total no. of Instructions = 100

No. of Instructions in MM = 80

MM's Hit Ratio = $80/100 = 0.8$ or 80%



Way 1:

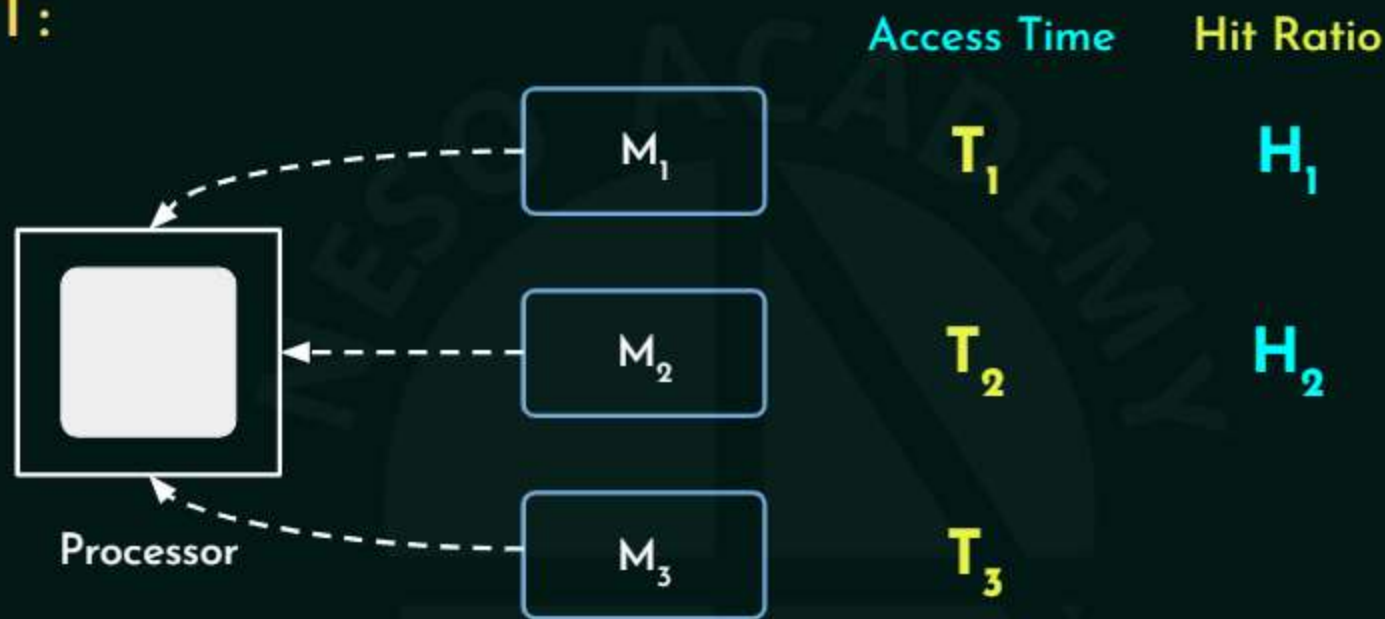


Effective/Average Memory Access Time (T_{avg}):

$$H_1 T_1 + ((1-H_1) \times H_2) T_2 + ((1-H_1) \times (1-H_2)) T_3$$



Way 1:

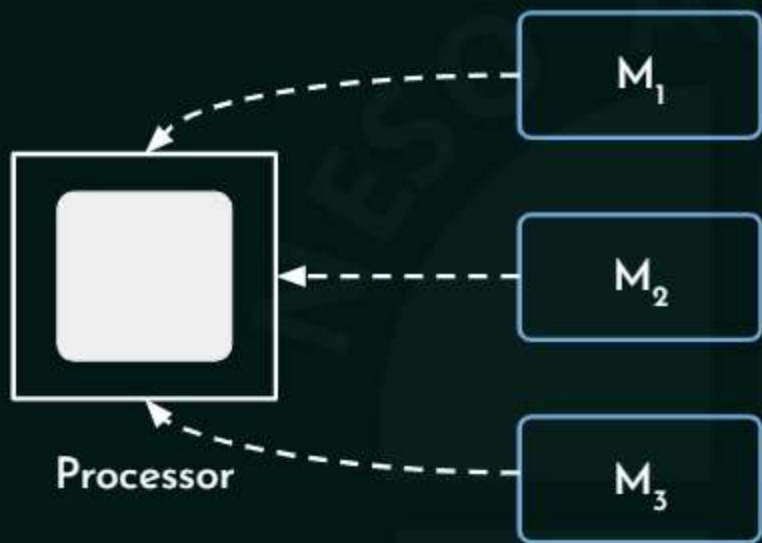


Effective/Average Memory Access Time (T_{avg}):

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Way 1:



Access Time

Hit Ratio

T_1

H_1

T_2

H_2

T_3

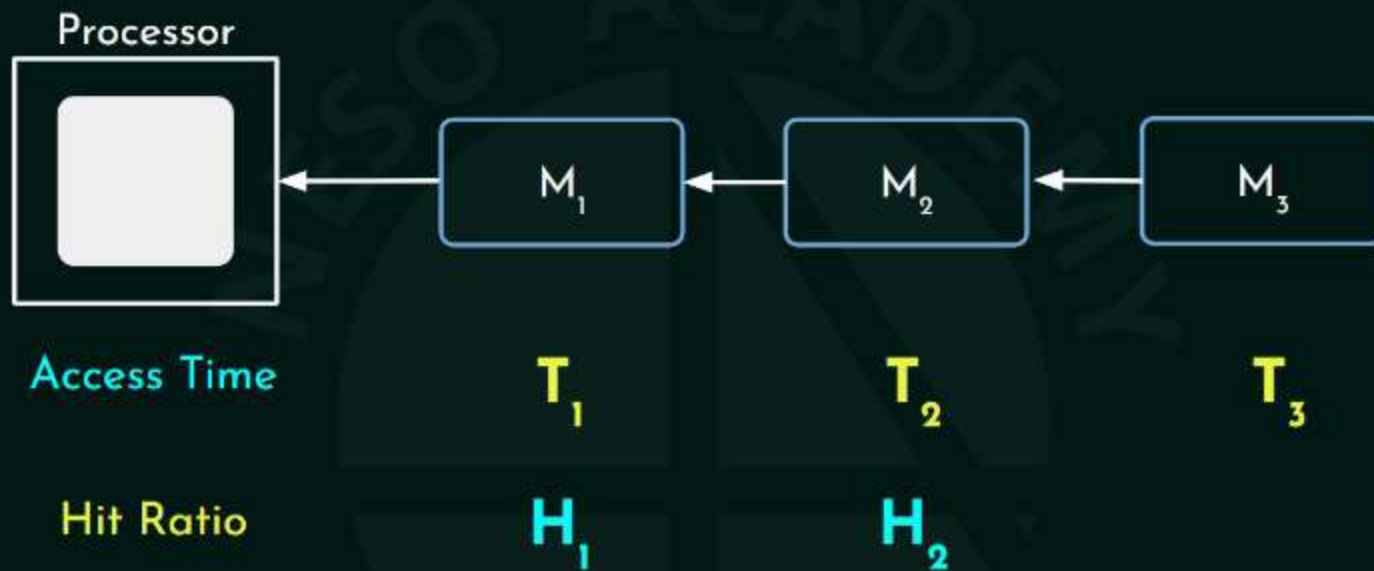
$(H_1 \times T_1)$

Time taken to
access M_1
if it's a hit in M_1





Way 2:



Effective/Average Memory Access Time (T_{avg}):

$$H_1 T_1 + ((1-H_1) \times H_2) (T_1 + T_2) + ((1-H_1) \times (1-H_2)) (T_1 + T_2 + T_3)$$



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Memory Interfacing – Solved PYQs

Q1: A cache memory needs an access time of 30 ns and main memory 150 ns, what is the average access time of CPU (assume hit ratio = 80%)?

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Sol.



Access Time

$$T_{\text{cache}} = 30 \text{ ns}$$

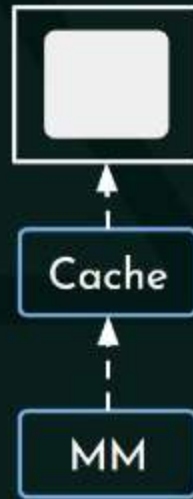
$$T_{\text{MM}} = 150 \text{ ns}$$

Hit Ratio

$$H_{\text{cache}} = 80\% = 0.8$$

$$\begin{aligned} T_{\text{avg}} &= H_{\text{cache}} T_{\text{cache}} + (1 - H_{\text{cache}}) T_{\text{MM}} \\ &= 0.8 \times 30 + (1 - 0.8) \times 150 \text{ ns} \\ &= 24 + 0.2 \times 150 = 54 \text{ ns} \end{aligned}$$

Processor



Access Time

$$T_{\text{cache}} = 30 \text{ ns}$$

$$T_{\text{MM}} = 150 \text{ ns}$$

Hit Ratio

$$H_{\text{cache}} = 80\% = 0.8$$

$$\begin{aligned} T_{\text{avg}} &= H_{\text{cache}} T_{\text{cache}} + (1 - H_{\text{cache}}) (T_{\text{cache}} + T_{\text{MM}}) \\ &= 0.8 \times 30 + (1 - 0.8) \times (30 + 150) \text{ ns} \\ &= 24 + 0.2 \times 180 = 60 \text{ ns} \end{aligned}$$

Q1: A cache memory needs an access time of 30 ns and main memory 150 ns, what is the average access time of CPU (assume hit ratio = 80%)?

ISRO CS 2017

- Sol. **A) 60**
B) 30
C) 150
D) 70

Processor



Cache

MM

Access Time

$$T_{\text{cache}} = 30 \text{ ns}$$

$$T_{\text{MM}} = 150 \text{ ns}$$

Hit Ratio

$$H_{\text{cache}} = 80\% = 0.8$$

$$\begin{aligned} T_{\text{avg}} &= H_{\text{cache}} T_{\text{cache}} + (1 - H_{\text{cache}}) (T_{\text{cache}} + T_{\text{MM}}) \\ &= 0.8 \times 30 + (1 - 0.8) \times (30 + 150) \text{ ns} \\ &= 24 + 0.2 \times 180 = 60 \text{ ns} \end{aligned}$$

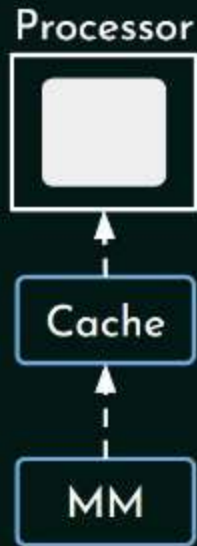
Q2: Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

GATE CS 2015

Q2: Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

GATE CS 2015

Sol.



Access Time

$$T_{\text{cache}} + T_{\text{MM}} = 50 \text{ ns}$$

$$T_{\text{cache}} = 5 \text{ ns}$$

Hit Ratio

$$H_{\text{cache}} = 80\% = 0.8$$

$$T_{\text{avg}} = H_{\text{cache}} T_{\text{cache}} + (1 - H_{\text{cache}}) (T_{\text{cache}} + T_{\text{MM}})$$

$$= 0.8 \times 5 + (1 - 0.8) \times 50 \text{ ns}$$

$$= 4 + 0.2 \times 50 = 14 \text{ ns}$$